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2027 CYCLE

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CUET UG

Mathematics

Algebra: Matrices & Determinants

Matrix Taxonomy, Algebraic Operations, Inverses, Determinant Power Laws & Linear Systems · Section A Unit 1 & Section B1 Unit II

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Aligned to the latest official syllabus · Concept-first explanations

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Syllabus Alignment & Exam Blueprint

Sub-Topic / Domain	Official NTA CUET UG Syllabus Scope (Mathematics 319)	Exam Cognitive Focus
Matrix Concept & Types	Concept, notation, order $m \times n$, equality of matrices; Types: Column, Row, Square, Diagonal, Scalar, Identity, Zero; Transpose of a matrix, Symmetric and Skew-symmetric matrices; Decomposition into symmetric and skew-symmetric parts.	Identifying matrix types on sight; extracting unknowns from matrix equality; applying skew-symmetric odd-order zero determinant rule; verifying $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$.
Matrix Operations & Inverses	Addition and scalar multiplication; Properties of addition and scalar multiplication; Matrix multiplication, non-commutativity ($AB \neq BA$), existence of zero divisors ($AB = O$ with $A, B \neq O$); Transpose reversal law $(AB)^T = B^T A^T$; Invertible matrices and uniqueness of inverse; Inversion reversal law $(AB)^{-1} = B^{-1}A^{-1}$.	Dimensional conformability for multiplication; calculating powers of matrices; distinguishing zero divisors from zero scalars; executing 2×2 and 3×3 matrix inverses.
Determinants & Minors/Cofactors	Determinant of a square matrix (up to 3×3); Minors and Cofactors; Cofactor expansion theorem; Orthogonal expansion identity ($\sum a_{ij}A_{kj} = 0$ for $i \neq k$); Multiplicative determinant laws: $\lvert AB\rvert = \lvert A\rvert\lvert B\rvert$, $\lvert kA\rvert = k^n \lvert A\rvert$, $\lvert A^T\rvert = \lvert A\rvert$, $\lvert A^{-1}\rvert = \frac{1}{\lvert A\rvert}$.	Evaluating 3×3 determinants rapidly; applying the scalar scaling power law k^n; identifying zero determinants from proportional rows; evaluating orthogonal cofactor sums.
Adjoint & Invertibility Theorems	Adjoint of a square matrix $\text{adj}(A) = \left[A_{ij}\right]^T$; Fundamental identity $A \cdot \text{adj}(A) = \text{adj}(A) \cdot A = \lvert A\rvert\, I_n$; Singular ($\lvert A\rvert = 0$) and Non-singular ($\lvert A\rvert \neq 0$) matrices; Adjoint power laws: $\lvert\text{adj}(A)\rvert = \lvert A\rvert^{n-1}$, $\lvert\text{adj}(\text{adj}(A))\rvert = \lvert A\rvert^{(n-1)^2}$, $\text{adj}(\text{adj}(A)) = \lvert A\rvert^{n-2} A$.	Using 2×2 lightning inverse shortcut; evaluating determinant power laws of nested adjoints without expanding elements.
Linear Systems & Geometry	Consistency, inconsistency, and number of solutions of systems of linear equations ($AX = B$); Solving systems with unique solutions via matrix inversion $X = A^{-1}B$; Cramer's rule; Applications in area of triangle and collinearity checks.	Classifying systems via $\lvert A\rvert$ and $(\text{adj}(A))B$; solving 2 and 3 variable systems; computing triangle areas and unknown vertex coordinates with $\pm$ area.

EXAM PATTERN

CUET UG Scoring Blueprint: Algebra (Matrices and Determinants) is universally mandatory across both the Common Section A (Unit 1) and the Domain Section B1 (Unit II) / Section B2 (Unit II). Candidates will encounter **5 to 8 questions (25 to 40 marks)** from this unit alone. Core NTA focus areas include: **determinant scalar power laws ($\lvert kA\rvert = k^n \lvert A\rvert$), adjoint determinant powers ($\lvert A\rvert^{n-1}$), inconsistency testing when $\lvert A\rvert = 0$**, and **reverse-order transposition/inversion laws**.

Matrix Foundations, Classification & Special Types

1. Mathematical Definition & Order

A **matrix** is an ordered rectangular array of numbers, symbols, or functions arranged in horizontal rows and vertical columns:

$$A = [a_{ij}]_{m\times n} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}$$

The matrix has m rows and n columns; its **order** (or dimension) is denoted by $m \times n$, containing exactly mn total elements.

- **Equality of Matrices:** Two matrices $A = [a_{ij}]$ and $B = [b_{ij}]$ are **equal** ($A = B$) if and only if:
  1. They have identical orders ($m_A = m_B$ and $n_A = n_B$).
  2. Every corresponding entry is equal: $a_{ij} = b_{ij}$ for all i, j.

2. Systematic Taxonomy of Matrix Types

- **Row Matrix:** Contains exactly one row: order $1 \times n$. (e.g., $[1 \quad 4 \quad 9]_{\{1\times 3\}}$).
- **Column Matrix:** Contains exactly one column: order $m \times 1$. (e.g., $[2; 5; 7]_{\{3\times 1\}}$).
- **Square Matrix:** Number of rows equals number of columns ($m = n$). Only square matrices possess determinants, traces, and inverses. The elements $a_{11}, a_{22}, ..., a_{nn}$ form the **principal diagonal**.
- **Diagonal Matrix:** A square matrix where all non-diagonal elements are zero: $a_{ij} = 0$ for all $i \neq j$.
- **Scalar Matrix:** A diagonal matrix where all diagonal entries are equal to a constant k: $a_{ii} = k$ and $a_{ij} = 0$ $(i \neq j)$.
- **Identity Matrix (I_n):** A scalar matrix with $k = 1$: $a_{ii} = 1$ and $a_{ij} = 0$ $(i \neq j)$. Satisfies $AI = IA = A$.
- **Zero (Null) Matrix (O):** Every entry is zero: $a_{ij} = 0$. Serves as the additive identity: $A + O = A$.

3. Transpose of a Matrix & Symmetry

The **transpose** of an $m \times n$ matrix A, denoted by A^T or A', is the $n \times m$ matrix obtained by interchanging rows and columns: $(A^T)_{ij} = a_{ji}$. **Fundamental Properties of Transpose:**

1. $(A^T)^T = A$
2. $(A + B)^T = A^T + B^T$
3. $(kA)^T = kA^T$ (for any scalar k)
4. **Reversal Law:** $(AB)^T = B^T A^T$ and $(ABC)^T = C^T B^T A^T$

Symmetric & Skew-Symmetric Matrices

Let A be a square matrix:

- **Symmetric Matrix:** $A^T = A$, meaning $a_{ij} = a_{ji}$ for all i, j. Entries are symmetric across the principal diagonal.
- **Skew-Symmetric Matrix:** $A^T = -A$, meaning $a_{ij} = -a_{ji}$ for all i, j.
  - **Diagonal Property:** For $i = j$, $a_{ii} = -a_{ii} \Rightarrow 2a_{ii} = 0 \Rightarrow a_{ii} = 0$. Every diagonal entry of a skew-symmetric matrix is strictly **zero**!
  - **Determinant Theorem (Odd Order):** If A is a skew-symmetric matrix of **odd order** ($n = 1, 3, 5, ...$), then $|A| = 0$ identically!

Proof: $|A| = |A^T| = |-A| = (-1)^n |A| = -|A| \Rightarrow 2|A| = 0 \Rightarrow |A| = 0$

Fundamental Decomposition Theorem

Any square matrix A can be **uniquely expressed** as the sum of a symmetric matrix and a skew-symmetric matrix:

$$A = \underbrace{\frac{1}{2}(A + A^T)}_{\text{Symmetric } P} + \underbrace{\frac{1}{2}(A - A^T)}_{\text{Skew-Symmetric } Q}$$

Where $P^T = \frac{1}{2}\left(A^T + (A^T)^T\right) = \frac{1}{2}(A^T + A) = P$, and $Q^T = \frac{1}{2}(A^T - A) = -\frac{1}{2}(A - A^T) = -Q$.

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Matrix Operations Engine & Inversion

1. Matrix Addition & Scalar Multiplication

- **Addition ($A+B$):** Defined if and only if A and B have the **exact same order** $m \times n$. Elements are added component-wise: $(A+B)_{ij} = a_{ij} + b_{ij}$.
  - Commutative: $A+B = B+A$ | Associative: $(A+B)+C = A+(B+C)$
  - Additive Identity: $A+O = O+A = A$ | Additive Inverse: $A+(-A) = O$
- **Scalar Multiplication (kA):** The scalar k multiplies **every single element** of the matrix: $(kA)_{ij} = ka_{ij}$.
  - $k(A+B) = kA+kB$ | $(k+l)A = kA+lA$ | $(kl)A = k(lA)$

2. Matrix Multiplication (AB)

Let A be of order $m \times n$ and B be of order $n \times p$. The product AB is defined if and only if the **number of columns of A equals the number of rows of B** (inner dimensions match):

$$(m \times n) \cdot (n \times p) \to (m \times p)$$

The entry c_{ij} in row i and column j of product $C = AB$ is the dot product of row i of A and column j of B:

$$c_{ij} = \sum_{k=1}^{n} a_{ik}b_{kj} = a_{i1}b_{1j} + a_{i2}b_{2j} + \ldots + a_{in}b_{nj}$$

Critical Properties & High-Yield Pitfalls

1. **Non-Commutative:** In general, $AB \neq BA$. Even if both AB and BA exist and have identical square orders, their entries usually differ completely.
2. **Associativity & Distributivity:** $(AB)C = A(BC)$ and $A(B+C) = AB+AC$.
3. **Multiplicative Identity:** For square matrix A of order n, $AI_n = I_nA = A$.
4. **Zero Divisors (Crucial CUET Trap):** In ordinary scalar arithmetic, $ab = 0 \Rightarrow a = 0$ or $b = 0$. **In matrix algebra, this is completely false!** Two non-zero matrices can multiply to yield the zero matrix ($AB = O$ with $A \neq O$ and $B \neq O$):

$$\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = O$$

5. **No Direct Cancellation:** $AB = AC$ does **not** imply $B = C$ unless A is non-singular ($|A| \neq 0$).

3. Special Square Matrix Classes

- **Idempotent Matrix:** $A^2 = A$. (Has eigenvalues 0 or 1; $\det(A) = 0$ or 1).
- **Involutory Matrix:** $A^2 = I$. (Its own inverse: $A^{-1} = A$; $\det(A) = \pm 1$).
- **Nilpotent Matrix:** $A^k = O$ for some positive integer k. (Index k; $\det(A) = 0$).
- **Orthogonal Matrix:** $AA^T = A^TA = I$. Implies $A^{-1} = A^T$ and $|A| = \pm 1$.

4. Invertible Matrices & The Reversal Law

A square matrix A of order n is said to be **invertible** if there exists a square matrix B of order n such that:

$$AB = BA = I_n$$

Then B is uniquely determined and called the **inverse of A**, written as $B = A^{-1}$.

- **Theorem of Uniqueness:** If a matrix inverse exists, it is strictly **unique**.
- **Reversal Law for Inverses:** Let A and B be invertible matrices of the same order. Then AB is invertible and:

$$(AB)^{-1} = B^{-1}A^{-1} \quad \text{and} \quad (ABC)^{-1} = C^{-1}B^{-1}A^{-1}$$

- **Transpose-Inverse Commutation:** $(A^T)^{-1} = (A^{-1})^T$.



Figure 1: Figure 1: High-yield NCERT analytical framework — Matrix taxonomy, algebraic operations engine, zero divisors, transpose reversal laws, and determinant multiplicative rules.

Key Analytical Takeaways from Figure 1:

1. **Matrix vs Determinant Scalar Scaling:** In matrix multiplication, kA multiplies **all mn elements**. In determinants, multiplying a single row by k multiplies the determinant by k. Therefore, for an $n \times n$ matrix, $|kA| = k^n\,|A|$! Forgetting the exponent n is the single most common student mistake in algebra.
2. **Skew-Symmetric Determinants:** The principal diagonal elements of any skew-symmetric matrix must be zero ($a_{ii} = 0$). For any odd dimension ($3 \times 3, 5 \times 5$), its determinant is identically zero.
3. **Adjoint Power Sequence:** For an $n \times n$ matrix, $|\text{adj}(A)| = |A|^{n-1}$ and $|\text{adj}(\text{adj}(A))| = |A|^{(n-1)^2}$. For $n = 3$, these simplify neatly to $|A|^2$ and $|A|^4$.

Determinants: Definitions, Expansion & Properties

1. Determinant Concept & Minors/Cofactors

To every square matrix $A = [a_{ij}]$ of order n, we associate a unique scalar number called the **determinant of A**, written as $|A|$ or $\det(A)$.

- **Order 1×1:** For $A = [a]$, $|A| = a$.
- **Order 2×2:** For $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, $|A| = ad - bc$.
- **Order 3×3:** Evaluated by expanding along any row or column:

$$|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix} = a_{11}\begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12}\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13}\begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

Minors (M_{ij}) and Cofactors (A_{ij})

- **Minor (M_{ij}):** The determinant of the submatrix of order $(n-1) \times (n-1)$ obtained by deleting row i and column j from matrix A.
- **Cofactor (A_{ij} or C_{ij}):** The signed minor defined by:

$$A_{ij} = (-1)^{i+j} M_{ij}$$

The signs follow a checkerboard pattern: $\begin{pmatrix} + & - & + \\ - & + & - \\ + & - & + \end{pmatrix}$.

Core Cofactor Expansion Theorems

1. **Expansion Along Row i:** The sum of products of elements of any row with their corresponding cofactors equals the determinant:

$$|A| = \sum_{j=1}^{n} a_{ij} A_{ij} = a_{i1}A_{i1} + a_{i2}A_{i2} + \ldots + a_{in}A_{in}$$

2. **Orthogonal Expansion Identity (Zero Sum Theorem):** If elements of any row are multiplied with the cofactors of **any other row**, their sum is identically **zero**:

$$\sum_{j=1}^{n} a_{ij} A_{kj} = a_{i1}A_{k1} + a_{i2}A_{k2} + \ldots + a_{in}A_{kn} = 0 \quad (\text{for } i \neq k)$$

(Similarly holds for column multiplications).

2. Fundamental Properties of Determinants

1. **Transpose Invariance:** Interchanging rows and columns leaves the determinant unchanged: $|A^T| = |A|$.
2. **Row/Column Swap:** If any two rows (or columns) are interchanged, the sign of the determinant is reversed: $\det(A') = -\det(A)$.
3. **Identical / Proportional Rows:** If any two rows or columns are identical or proportional, $|A| = 0$.
4. **Scalar Factor Law:** If every element of a row or column is multiplied by k, the determinant is multiplied by k. For an $n \times n$ matrix:

$$|kA| = k^n \ |A|$$

5. **Elementary Operations:** The operation $R_i \to R_i + kR_j$ (or $C_i \to C_i + kC_j$) leaves the value of the determinant **completely unchanged**.
6. **Product Rule:** For square matrices of the same order:

$$|AB| = |A| \cdot |B| \quad \text{and} \quad |A^{-1}| = \frac{1}{|A|}$$

7. **Triangular Matrices:** If A is upper triangular, lower triangular, or diagonal, its determinant equals the **product of its diagonal entries**: $|A| = a_{11}a_{22}\ldots a_{nn}$.

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Adjoint, Inverses & Matrix Equation Solvers

1. Adjoint of a Matrix (adj(A))

The **adjoint** of a square matrix $A = [a_{ij}]$, denoted by adj(A), is formally defined as the **transpose of the matrix of cofactors** of A:

$$\text{adj}(A) = [A_{ij}]^T = \begin{pmatrix} A_{11} & A_{21} & \dots & A_{n1} \\ A_{12} & A_{22} & \dots & A_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ A_{1n} & A_{2n} & \dots & A_{nn} \end{pmatrix}$$

High-Yield Warning: Note that A_{12} sits in row 2, column 1 because the cofactor matrix has been transposed!

The Fundamental Adjoint Identity

For any square matrix A of order n:

$$A \cdot \text{adj}(A) = \text{adj}(A) \cdot A = |A| \, I_n$$

- **Singular vs Non-Singular:** Matrix A is **singular** if $|A| = 0$; it is **non-singular** if $|A| \neq 0$.
- **Formula for Matrix Inverse:** Dividing the fundamental identity by $|A|$ (when $|A| \neq 0$) yields:

$$A^{-1} = \frac{1}{|A|} \text{adj}(A)$$

Invertibility Criterion: A square matrix A possesses an inverse **if and only if it is non-singular ($|A| \neq 0$)**.

Master Adjoint Power Laws (Order n)

- Determinant of Adjoint: $|\text{adj}(A)| = |A|^{n-1}$
- Determinant of Double Adjoint: $|\text{adj}(\text{adj}(A))| = |A|^{(n-1)^2}$
- Double Adjoint Matrix: $\text{adj}(\text{adj}(A)) = |A|^{n-2} \, A$
- Product of Adjoints: $\text{adj}(AB) = \text{adj}(B) \cdot \text{adj}(A)$
- Scalar with Adjoint: $\text{adj}(kA) = k^{n-1} \, \text{adj}(A)$

The 2×2 Lightning Inverse Shortcut

For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ with $|A| = ad - bc \neq 0$:

1. **Swap the main diagonal entries:** $a \leftrightarrow d$.
2. **Negate the anti-diagonal entries:** $b \to -b$, $c \to -c$.
3. **Divide by determinant:**

$$A^{-1} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

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Systems of Linear Equations & Consistency Criteria

1. Matrix Representation of Linear Systems

Consider a non-homogeneous system of three linear equations in three variables:

$$a_1x + b_1y + c_1z = d_1$$
$$a_2x + b_2y + c_2z = d_2$$
$$a_3x + b_3y + c_3z = d_3$$

This system can be written compactly as the single matrix equation:

$$AX = B$$

where:

$$A = \begin{pmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{pmatrix}, \quad X = \begin{pmatrix} x \\ y \\ z \end{pmatrix}, \quad B = \begin{pmatrix} d_1 \\ d_2 \\ d_3 \end{pmatrix}$$

Here A is the **coefficient matrix**, X is the **variable vector**, and B is the **constant vector**.

2. Consistency Analysis Algorithm (The Matrix Method)

To determine the existence and uniqueness of solutions, evaluate the following decision tree:

Condition 1	Condition 2	Consistency Classification	Nature of Solutions & Geometric Meaning
$\|A\| \neq 0$	Any B	**Consistent & Independent**	Unique solution: $X = A^{-1}B$. Three planes intersect at a single unique point.
$\|A\| = 0$	$(\text{adj}(A))B \neq O$	**Inconsistent**	Zero solutions (No solution). Planes are parallel or intersect in pairs along disjoint lines.
$\|A\| = 0$	$(\text{adj}(A))B = O$	**Consistent & Dependent**	Infinitely many solutions (or inconsistent if planes are parallel). Planes coincide or share a line.

3. Homogeneous Systems ($AX = O$)

When the constant vector $B = O$, the system takes the form $AX = O$:

$$a_1x + b_1y + c_1z = 0, \quad a_2x + b_2y + c_2z = 0, \quad a_3x + b_3y + c_3z = 0$$

- The **trivial solution** $x = 0, y = 0, z = 0$ $(X = O)$ is always a solution. Hence, homogeneous systems are **always consistent**.
- **Condition for Unique Solution:**
$$|A| \neq 0 \iff \text{Only the trivial solution } X = O \text{ exists.}$$
- **Condition for Non-Trivial Solutions:**
$$|A| = 0 \iff \text{Infinitely many non-trivial (non-zero) solutions exist.}$$

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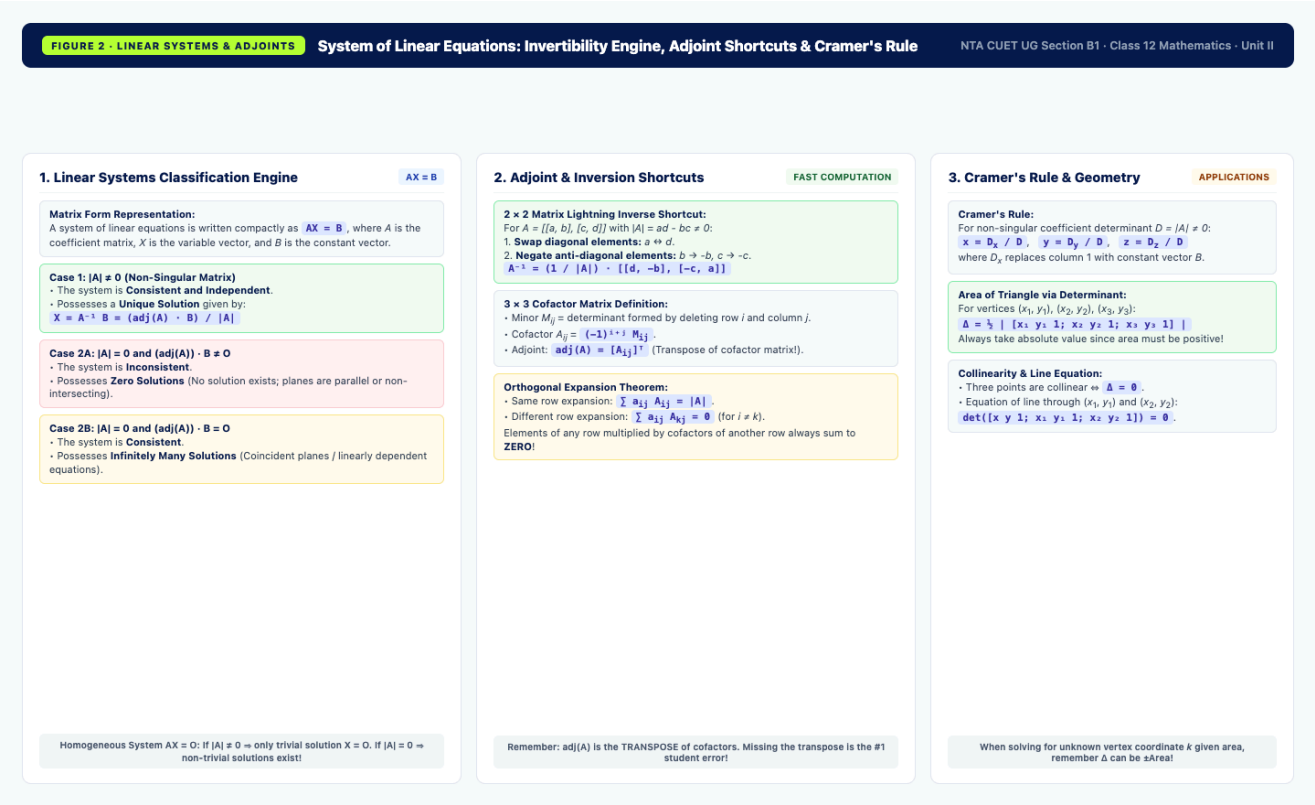


Figure 2: Figure 2: Systematic comparison matrix — Linear systems consistency flowchart ($AX = B$), adjoint shortcuts, 2×2 lightning inverse, and geometric applications via determinants.

Key Analytical Takeaways from Figure 2:

1. **Consistency Branching:** When $|A| \neq 0$, the system is guaranteed to have a unique solution given by $X = A^{-1}B$. When $|A| = 0$, you **cannot declare no solution immediately** — you must compute the matrix product $(\text{adj}(A))B$. If $(\text{adj}(A))B = O$, infinitely many solutions exist!
2. **Homogeneous Systems ($AX = O$):** Because the constant vector is O, the trivial solution $x = y = z = 0$ always exists. Non-trivial (non-zero) solutions exist **if and only if $|A| = 0$**.
3. **Cramer's Rule Engine:** In systems where $|A| \neq 0$, each variable x_i is computed independently by replacing column i of the coefficient matrix with the constant vector B: $x = \frac{D_x}{D}, y = \frac{D_y}{D}, z = \frac{D_z}{D}$.

Advanced Matrix Applications & Geometric Toolkit

1. Cayley-Hamilton Theorem & Matrix Polynomials

Cayley-Hamilton Theorem: Every square matrix A satisfies its own characteristic equation:

$$\det(A - \lambda I) = 0$$

- For a 2×2 matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the characteristic equation is:

$$\lambda^2 - \text{Tr}(A)\lambda + |A| = 0 \Rightarrow A^2 - (a+d)A + (ad - bc)I = O$$

- **Computing Inverses via Matrix Polynomials:** Multiplying $A^2 - \text{Tr}(A)A + |A|\ I = O$ by A^{-1} yields:

$$A - \text{Tr}(A)I + |A|\ A^{-1} = O \Rightarrow A^{-1} = \frac{1}{|A|}(\text{Tr}(A)I - A)$$

  This provides an instantaneous algebraic route to find inverses and higher matrix powers (A^3, A^4) without manual matrix multiplication.

2. Geometric Applications: Area of Triangle & Collinearity

- **Area of Triangle:** The area of a triangle with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ is:

$$\Delta = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

  Because area is a positive physical quantity, always take the absolute value $|\Delta|$. When solving for an unknown coordinate given a known area, remember that the determinant can equal + Area or − Area.

- **Collinearity Condition:** Three points are collinear if and only if $\Delta = 0$.

- **Line Equation:** The equation of the line passing through (x_1, y_1) and (x_2, y_2) is:

$$\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

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Key Facts & Terms Cheatsheet

MASTER DIAGNOSTIC & CONCEPTUAL CHEATSHEET — MATRICES & DETERMINANTS

Key Mathematical Topic	Core NCERT Mechanism / Definition	Exam Identifier
Square Matrix	$m = n$. Only square matrices have determinants, trace, and inverses.	Rows = Columns
Diagonal Matrix	$a_{ij} = 0$ for all $i \neq j$. Non-zero entries exist only on main diagonal.	Off-Diagonal = 0
Identity Matrix I_n	Diagonal entries all 1, off-diagonal 0. Multiplicative identity: $AI = IA = A$.	Diagonal = 1
Symmetric Matrix	$A^T = A$, meaning $a_{ij} = a_{ji}$ mirror symmetry across main diagonal.	$A^T = A$
Skew-Symmetric Matrix	$A^T = -A$; $a_{ii} = 0$ always. Odd order skew-symmetric has $\|A\| = 0$.	$a_{ii} = 0, \|A\| = 0$
Matrix Decomposition	Every square matrix is uniquely $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$.	Symm + Skew-Symm
Matrix Product AB	Conformable if columns of A equal rows of B: $(m \times n)(n \times p) \to m \times p$.	Inner Match
Non-Commutativity	In general, $AB \neq BA$. Matrix multiplication is strictly non-commutative.	$AB \neq BA$
Zero Divisors	$AB = O$ does NOT imply $A = O$ or $B = O$. Non-zero matrices have zero product.	Zero Product
Transpose Reversal	$(AB)^T = B^T A^T$ and $(ABC)^T = C^T B^T A^T$.	Order Reverses
Inverse Reversal	$(AB)^{-1} = B^{-1} A^{-1}$ (only for non-singular square matrices).	Order Reverses
Determinant Product	$\|AB\| = \|A\| \cdot \|B\|$ and $\|A^k\| = \|A\|^k$.	$\|AB\| = \|A\|\|B\|$
Scalar Scaling Law	$\|kA\| = k^n \|A\|$ for an $n \times n$ matrix (each row contributes k).	Factor k^n
Transpose Determinant	$\|A^T\| = \|A\|$ (rows and columns interchanged leaves det invariant).	$\|A^T\| = \|A\|$
Inverse Determinant	$\|A^{-1}\| = \frac{1}{\|A\|} = (\|A\|)^{-1}$.	Reciprocal
Adjoint Identity	$A \cdot \text{adj}(A) = \text{adj}(A) \cdot A = \|A\| \, I_n$.	$\|A\| \, I_n$
Adjoint Determinant	$\|\text{adj}(A)\| = \|A\|^{n-1}$ for matrix of order n. For $n = 3$, $\|A\|^2$.	$\|A\|^{n-1}$
Double Adjoint Det	$\|\text{adj}(\text{adj}(A))\| = \|A\|^{(n-1)^2}$. For $n = 3$, $\|A\|^4$.	$\|A\|^{(n-1)^2}$
2×2 Inverse Shortcut	Swap diagonal entries, negate anti-diagonal, divide by $\|A\|$.	Swap & Negate
Orthogonal Expansion	$\sum a_{ij} A_{kj} = 0$ when elements of row i multiply cofactors of row $k \neq i$.	Orthogonal = 0
Linear System $AX = B$	Unique solution $X = A^{-1}B$ if $\|A\| \neq 0$. Inconsistent if $\|A\| = 0$ & $(\text{adj } A)B \neq O$.	Consistency Tree
Area of Triangle	$\Delta = \frac{1}{2} \|[x_1 y_1 1; x_2 y_2 1; x_3 y_3 1]\|$. Collinear if $\Delta = 0$.	Determinant Area

Common Errors and High-Yield CUET Traps

COMMON STUDENT MISTAKES & DIAGNOSTIC TRAPS — MATRICES & DETERMINANTS

Common Student Mistake	Scientifically Verified Reality	The Exact Trap Mechanism
Confusing $\|kA\|$ with $k\|A\|$	For an $n \times n$ matrix, $\|kA\| = k^n \|A\|$ because k is factored out of **every single row** of the determinant.	For a 3×3 matrix with $\|A\| = 5$, students write $\|2A\| = 2(5) = 10$ instead of $2^3(5) = 40$.
Assuming $AB = O \Rightarrow A = O$ or $B = O$	Matrix algebra contains zero divisors: two non-zero matrices can multiply to give the null matrix O.	Questions ask "If $AB = O$, which must be true?" Options include $A = O$ or $B = O$; students pick it mistakenly.
Forgetting Transpose on Adjoint	$\text{adj}(A)$ is the **transpose** of the cofactor matrix: $\text{adj}(A) = [A_{ij}]^T$. It is NOT just $[A_{ij}]$.	Students calculate minors and cofactors correctly but forget to transpose the matrix, swapping A_{12} and A_{21}.
Failing to Check $\|A\| \neq 0$ Before Inverting	A matrix is invertible if and only if it is non-singular ($\|A\| \neq 0$). Singular matrices have no inverse.	Students algebraically compute $\text{adj}(A)$ for a singular matrix and forget that dividing by $\|A\| = 0$ is undefined.
Cancelling A from $AB = AC$	$AB = AC$ implies $B = C$ **only if A is non-singular ($\|A\| \neq 0$)**. If A is singular, B and C need not be equal.	NTA provides $B = C$ as a distractor without stating whether $\|A\| \neq 0$.
Declaring No Solution Whenever $\|A\| = 0$	If $\|A\| = 0$, the system could be **inconsistent** (no solution) OR **consistent** (infinitely many solutions if $(\text{adj } A)B = O$).	Students assume $\|A\| = 0$ automatically means zero solutions without calculating $(\text{adj}(A))B$.
Single Sign in Triangle Area Unknowns	When given area $= 35$, the determinant $\|\Delta\|$ can equal $+70$ or -70, yielding **two distinct coordinates for k**.	Students set $\Delta = +35$ only, finding only one value of k and missing the second valid coordinate in options.
Evaluating $AB \neq BA$ as Always False	While multiplication is non-commutative in general, commuting pairs **do exist** (e.g. $AI = IA$, $AA^{-1} = A^{-1}A$, powers A^pA^q).	Questions ask if $AB = BA$ is possible; students declare it impossible everywhere, forgetting diagonal/identity cases.

OurMocks

The Whole Chapter on One Page

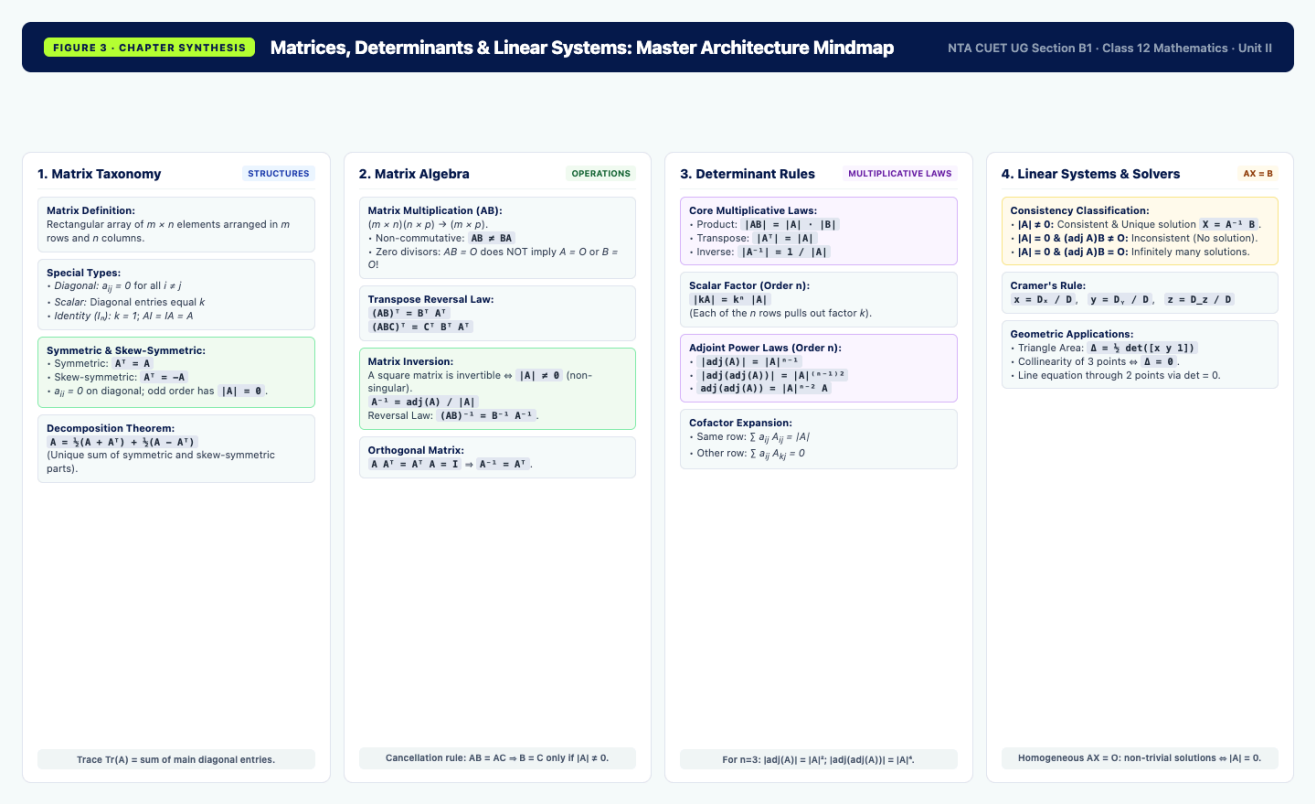


Figure 3: Figure 3: Master synthesis mindmap — high-yield integrated architecture of Matrices, Determinants, Inverses, Adjoint power laws, and Linear Systems ($AX = B$).

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Practice Questions — Part 1 (Q1 to Q5)

Q1. If matrix A has order 3×4, matrix B has order 4×2, and matrix C has order 2×3, then the order of the transpose matrix $((AB)C)^T$ is:

(A) 3×3 **(B)** 4×4
(C) 2×3 **(D)** 3×2

Sol. Q1. We check the dimension conformability step-by-step:

1. Product AB: $(3 \times 4) \cdot (4 \times 2) \to (3 \times 2)$.
2. Product $(AB)C$: $(3 \times 2) \cdot (2 \times 3) \to (3 \times 3)$.
3. Transpose $((AB)C)^T$: The transpose of an $m \times n$ matrix is $n \times m$.

For a 3×3 matrix, the transpose has order 3×3. **Answer: (A)**

Q2. If A is any arbitrary square matrix, then the matrix $A - A^T$ is always:

(A) A symmetric matrix **(B)** A skew-symmetric matrix
(C) An orthogonal matrix **(D)** A scalar matrix

Sol. Q2. Let $M = A - A^T$. We test for symmetry by taking the transpose of M:

$$M^T = (A - A^T)^T = A^T - (A^T)^T = A^T - A = -(A - A^T) = -M$$

Since $M^T = -M$, the matrix $A - A^T$ is strictly a **skew-symmetric matrix** for any square matrix A. (Note: $A + A^T$ is always symmetric because $(A + A^T)^T = A^T + A = A + A^T$). **Answer: (B)**

Q3. Consider the matrices $A = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix}$. Which of the following statements is correct regarding their product?

(A) $AB \neq O$ and $BA \neq O$ **(B)** $AB = O$ and $BA = O$
(C) $AB = O$ but $BA \neq O$ **(D)** $AB \neq O$ but $BA = O$

Sol. Q3. We compute both products directly:

1. $AB = \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0(3)+2(0) & 0(0)+2(0) \\ 0(3)+0(0) & 0(0)+0(0) \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} = O.$
2. $BA = \begin{pmatrix} 3 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 3(0)+0(0) & 3(2)+0(0) \\ 0(0)+0(0) & 0(2)+0(0) \end{pmatrix} = \begin{pmatrix} 0 & 6 \\ 0 & 0 \end{pmatrix} \neq O.$

Thus, $AB = O$ but $BA \neq O$. This demonstrates both the **existence of zero divisors** and the **non-commutativity of matrix multiplication**. **Answer: (C)**

Q4. If A is a square matrix of order 3×3 such that $|A| = 5$, then the value of $|2A|$ is equal to:

(A) 10 **(B)** 20
(C) 40 **(D)** 80

Sol. Q4. Recall the determinant scalar scaling theorem for an $n \times n$ matrix:

$$|kA| = k^n \ |A|$$

Here $k = 2$, order $n = 3$, and $|A| = 5$. Substituting these values:

$$|2A| = 2^3 \cdot |A| = 8 \cdot 5 = 40$$

(Trap: Students who multiply k by $|A|$ directly obtain $2 \times 5 = 10$, which is Option A). **Answer: (C)**

Q5. If A is a non-singular square matrix of order 3×3 with $|A| = 4$, then the values of $|\text{adj}(A)|$ and $|\text{adj}(\text{adj}(A))|$ are respectively:

(A) 16 and 64 **(B)** 16 and 256
(C) 4 and 16 **(D)** 64 and 256

Sol. Q5. We apply the master adjoint power laws for order $n = 3$:

1. $|\text{adj}(A)| = |A|^{n-1} = |A|^{3-1} = |A|^2 = 4^2 = 16.$
2. $|\text{adj}(\text{adj}(A))| = |A|^{(n-1)^2} = |A|^{(3-1)^2} = |A|^{2^2} = |A|^4 = 4^4 = 256.$

Thus, the values are 16 and 256 respectively. **Answer: (B)**

Practice Questions — Part 2 (Q6 to Q10)

Q6. The inverse of the 2×2 matrix $A = \begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$ is equal to:

(A) $\begin{pmatrix} \frac{4}{5} & -\frac{3}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{pmatrix}$ **(B)** $\begin{pmatrix} \frac{4}{5} & \frac{3}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{pmatrix}$

(C) $\begin{pmatrix} -\frac{4}{5} & \frac{3}{5} \\ \frac{1}{5} & -\frac{2}{5} \end{pmatrix}$ **(D)** $\begin{pmatrix} \frac{2}{5} & -\frac{3}{5} \\ -\frac{1}{5} & \frac{4}{5} \end{pmatrix}$

Sol. Q6. We use the 2×2 lightning inverse shortcut:

1. Determinant: $|A| = (2)(4) - (3)(1) = 8 - 3 = 5 \neq 0$.
2. Form adjoint by swapping diagonal entries ($2 \leftrightarrow 4$) and negating anti-diagonal entries ($3 \to -3, 1 \to -1$):
$$\text{adj}(A) = \begin{pmatrix} 4 & -3 \\ -1 & 2 \end{pmatrix}$$
3. Inverse $A^{-1} = \frac{1}{|A|}\,\text{adj}(A) = \frac{1}{5}\begin{pmatrix} 4 & -3 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} \frac{4}{5} & -\frac{3}{5} \\ -\frac{1}{5} & \frac{2}{5} \end{pmatrix}$.

Answer: (A)

Q7. If $A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$, then the matrix product $A \cdot \text{adj}(A)$ is equal to:

(A) $\begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$ **(B)** $\begin{pmatrix} 5 & 0 \\ 0 & 5 \end{pmatrix}$

(C) $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ **(D)** $\begin{pmatrix} 7 & 1 \\ -1 & 7 \end{pmatrix}$

Sol. Q7. By the Fundamental Adjoint Identity:
$$A \cdot \text{adj}(A) = |A|\, I_n$$
For $A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$, determinant $|A| = (3)(2) - (1)(-1) = 6 + 1 = 7$. Therefore:
$$A \cdot \text{adj}(A) = 7I_2 = 7\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 7 & 0 \\ 0 & 7 \end{pmatrix}$$

Answer: (A)

Q8. Let $A = [a_{ij}]$ be a 3×3 matrix and A_{ij} denote the cofactor of element a_{ij}. The numerical value of the expression $a_{11}A_{21} + a_{12}A_{22} + a_{13}A_{23}$ is:

(A) $|A|$ **(B)** $-|A|$

(C) 0 **(D)** $|A|^2$

Sol. Q8. By the **Orthogonal Expansion Theorem**:

- If elements of any row are multiplied with their **own corresponding cofactors**, the sum equals the determinant: $\sum a_{1j}A_{1j} = |A|$.
- If elements of any row are multiplied with the cofactors of **any other row**, the sum is identically zero: $\sum a_{1j}A_{2j} = 0$.

Here, the elements belong to Row 1, but the cofactors belong to Row 2 ($1 \neq 2$). Hence, the sum is strictly 0. **Answer: (C)**

Q9. If A is a skew-symmetric matrix of order 5×5, then the determinant $|A|$ is equal to:

(A) 5 **(B)** -1

(C) 1 **(D)** 0

Sol. Q9. By definition of skew-symmetric matrix, $A^T = -A$. Taking the determinant of both sides:
$$|A^T| = |-A|$$
Since $|A^T| = |A|$ and for order $n = 5$, $|-A| = (-1)^5\,|A| = -|A|$:
$$|A| = -|A| \Rightarrow 2\,|A| = 0 \Rightarrow |A| = 0$$
Thus, the determinant of any skew-symmetric matrix of odd order is identically zero. **Answer: (D)**

Q10. If the area of a triangle with vertices $(2, -6)$, $(5, 4)$, and $(k, 4)$ is 35 sq units, then the possible values of k are:

(A) 12 only **(B)** -2 only

(C) $12, -2$ **(D)** $-12, 2$

Sol. Q10. The determinant formula for triangle area is:
$$\Delta = \frac{1}{2}\begin{vmatrix} 2 & -6 & 1 \\ 5 & 4 & 1 \\ k & 4 & 1 \end{vmatrix} = \pm 35 \Rightarrow \begin{vmatrix} 2 & -6 & 1 \\ 5 & 4 & 1 \\ k & 4 & 1 \end{vmatrix} = \pm 70$$
Expand along Row 1:
$$2(4-4) - (-6)(5-k) + 1(20-4k) = 0 + 6(5-k) + 20 - 4k = 30 - 6k + 20 - 4k = 50 - 10k$$
Now solve both cases:

1. $50 - 10k = 70 \Rightarrow -10k = 20 \Rightarrow k = -2$
2. $50 - 10k = -70 \Rightarrow -10k = -120 \Rightarrow k = 12$

Thus, $k = 12$ or -2. (NTA trap: students who forget the -35 branch find only $k = -2$). **Answer: (C)**

Practice Questions — Part 3 (Q11 to Q15)

Q11. The values of x for which the matrix $A = \begin{pmatrix} x+1 & -2 \\ 3 & x-4 \end{pmatrix}$ is singular are:

(A) $1, 2$ (B) $-1, 4$
(C) $2, 3$ (D) $-2, 1$

Sol. Q11. A square matrix is singular if and only if its determinant is zero ($|A| = 0$):

$$|A| = (x+1)(x-4) - (-2)(3) = x^2 - 3x - 4 + 6 = x^2 - 3x + 2 = 0$$

Factoring the quadratic equation:

$$(x-1)(x-2) = 0 \Rightarrow x = 1 \quad \text{or} \quad x = 2$$

Answer: (A)

Q12. The system of linear equations $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + \lambda z = \mu$ has a unique solution if and only if:

(A) $\lambda = 3, \mu = 10$ (B) $\lambda \neq 3$, for any value of μ
(C) $\lambda = 3, \mu \neq 10$ (D) $\lambda \neq 3, \mu \neq 10$

Sol. Q12. By the Matrix Method, a system $AX = B$ has a unique solution if and only if the coefficient determinant $|A| \neq 0$:

$$|A| = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{vmatrix} = 1(2\lambda - 6) - 1(\lambda - 3) + 1(2 - 2) = 2\lambda - 6 - \lambda + 3 = \lambda - 3$$

For a unique solution:

$$|A| \neq 0 \Rightarrow \lambda - 3 \neq 0 \Rightarrow \lambda \neq 3$$

This condition depends solely on the coefficient matrix and holds for **any real value of μ**. **Answer: (B)**

Q13. If A and B are invertible matrices such that $A^{-1} = \begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix}$ and $B^{-1} = \begin{pmatrix} -1 & 1 \\ 2 & 0 \end{pmatrix}$, then the inverse matrix $(AB)^{-1}$ is:

(A) $\begin{pmatrix} -1 & 5 \\ 2 & -4 \end{pmatrix}$ (B) $\begin{pmatrix} -1 & -2 \\ 6 & 0 \end{pmatrix}$
(C) $\begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix}$ (D) $\begin{pmatrix} 6 & 0 \\ 5 & -1 \\ -4 & 2 \end{pmatrix}$

Sol. Q13. We apply the Reversal Law of Matrix Inverses:

$$(AB)^{-1} = B^{-1}A^{-1}$$

Substitute the given inverse matrices in exact reverse order:

$$(AB)^{-1} = \begin{pmatrix} -1 & 1 \\ 2 & 0 \end{pmatrix}\begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} (-1)(1) + (1)(0) & (-1)(-2) + (1)(3) \\ (2)(1) + (0)(0) & (2)(-2) + (0)(3) \end{pmatrix} = \begin{pmatrix} -1 & 5 \\ 2 & -4 \end{pmatrix}$$

(Trap: Multiplying $A^{-1}B^{-1}$ produces Option B). **Answer: (A)**

Q14. If matrix $A = \begin{pmatrix} 3 & 1 \\ -1 & 2 \end{pmatrix}$ satisfies the polynomial relation $A^2 - 5A + 7I = O$, then A^{-1} expressed in terms of A and I is:

(A) $A^{-1} = \frac{1}{7}(5I - A)$ (B) $A^{-1} = \frac{1}{7}(A - 5I)$
(C) $A^{-1} = 5I - A$ (D) $A^{-1} = \frac{1}{5}(7I - A)$

Sol. Q14. Multiply the given polynomial equation by A^{-1} from the left:

$$A^{-1}(A^2 - 5A + 7I) = A^{-1}O \Rightarrow A - 5I + 7A^{-1} = O$$

Isolate $7A^{-1}$:

$$7A^{-1} = 5I - A \Rightarrow A^{-1} = \frac{1}{7}(5I - A)$$

Answer: (A)

Q15. Let A be a 3×3 square matrix with $|A| = 2$. The numerical value of the determinant $|3 \cdot \text{adj}(2A)|$ is equal to:

(A) 6912 (B) 2304
(C) 768 (D) 576

Sol. Q15. We break the expression down using determinant and adjoint power laws:

1. For order $n = 3$, matrix $M = \text{adj}(2A)$ is 3×3. Therefore:
$$|3M| = 3^3 \ |M| = 27 \ |\text{adj}(2A)|$$
2. Using $|\text{adj}(B)| = |B|^{n-1} = |B|^2$ for $B = 2A$:
$$|\text{adj}(2A)| = (|2A|)^2$$
3. Calculate $|2A|$: $|2A| = 2^3 \ |A| = 8(2) = 16$.
4. Then $|\text{adj}(2A)| = (16)^2 = 256$.
5. Multiplying: $|3 \cdot \text{adj}(2A)| = 27 \cdot 256 = 6912$.

Answer: (A)